Yard Operation Optimisation

Oct 15, 2026

Problem Statements

This document outlines the pain points Cogent raised as concrete problem statements and actionables so that we can confirm we are targeting the right problems. Each problem statement is followed by a proposal, how we would measure success, the limitations and key enablers for the project to be implemented.

These are preliminary assessments of current conditions, where more changes and refinements to the plan, scope, priorities along the way are to be expected. We would appreciate any confirmation or correction to this in writing, or we can go through it together in another meeting.

Problem as we understand it	What we propose	Where we are
Weekend reshuffling and weekday relocations take crane/forklift and manpower time	A smarter placement of arriving containers on weekdays and a more efficient weekend plan	Weekend planning prototype tested on simulated bays
Container location is not recorded exactly, so time and effort is lost searching	System-assigned slots, confirmed by the operator, shown on a live yard map	Placement method planned; tracking to be designed with you
Instructions by phone or paper cause delays or mistakes	An integrated job list between office, operators and drivers	To be confirmed
Staff spend time answering routine queries from client and requests from other departments	A form/portal that can provide answers from the database	To be confirmed

1: Reduce manpower needed for yard moves

Our understanding. Weekend reshuffling and weekday relocations (moving containers that block a pickup) take crane and forklift time that is hard to staff. Operators must spend hours on the weekend to reorganise the yard.

What we propose.

- **Weekend plan:** a list of moves that minimises weekend work. We have a working prototype/demo on simulated bays.
- **Placement of arriving containers (next step):** a suggested slot for each arriving container, so fewer containers end up on top of one that leaves earlier, saving overtime/weekend work.

How we would measure success (baseline measured first, targets agreed with you):

- Crane and forklift overtime or hours on weekends.

Limitations.

- The plan can only use pickup dates known at the weekend. Late or changed bookings will still cause relocations.
- Whether weekend moves pay off depends on what a weekday relocation costs compared with a weekend move.
- Results so far come from simulated bays, not your yard. The model covers one bay at a time, with one height limit and one crane.
- A plan is only as good as the recorded container positions (see Aim 2).

Key enablers.

- Yard layout and few months of historical data (dwell time, pickup date) so we can model the system and demand faithfully.
- How early pickup dates are usually known, and how often they change.
- Crew size and hours spent on weekdays and weekends, and roughly what a weekend hour costs compared with a weekday hour.
- Height limits and typical fill ratio for each part of the yard, which equipment serves which area, and roughly how long one move takes.

2: Know the exact location of every container

Our understanding. Container positions are not always recorded correctly. Time and effort are both lost searching manually, and any plan built on a wrong position fails.

What we propose.

- The system suggests a slot for each arriving container and each move. The office and the operator see the same suggestion and agree on it.
- The operator confirms the actual placement with a tap or scan on a tablet in the cab. An override is allowed, and recorded.
- The office sees a live yard map built from these confirmations.

How we would measure success (baseline measured first, targets agreed with you):

- Turnaround time
- Weekend hour saving from a more efficient weekday placement
- Share of containers found exactly where the record says, checked by a spot audit.
- Staff time spent searching for containers per week.

Limitations.

- The system knows only what is confirmed. A container placed elsewhere without confirmation makes the map wrong.
- Positions already recorded wrongly today need a one-time yard check before go-live.
- It does not identify containers automatically. Camera or RFID reading would be a separate, later step.

Key enablers.

- One-time yard thorough survey before the pilot starts.
- Access to system or spreadsheet holding positions today.
- Agreement on how operators override a suggested slot, for example when it is blocked.

3: Improve communication between office, operators and drivers (to be confirmed)

Our understanding (to be confirmed). Instructions pass by phone or paper. This may cause delays, repeated calls or wrong moves.

What we propose.

- A job list showing who moves which container, from where, to where, and in what order.
- The queue of arriving trucks, visible to the office and operators.
- A job is marked done when the operator confirms the move (Aim 2), so the office does not need to call to check.

How we would measure success (baseline measured first, targets agreed with you):

- Wrong or repeated moves per week.
- Time from a job being issued to it being done.
- Phone calls/messages needed per job, from a short sample.

Limitations.

- It only helps if it is quicker to use than the phone or paper. It must be simple and work in the cab, possibly in more than one language.
- It depends on the location tracking in Aim 2.

Key enablers.

- A description of who talks to whom, about what, and through which channel.
- A few operators and drivers willing to try early versions and tell us what gets in the way.

4: Answer client and department queries faster (to be confirmed)

Our understanding (to be confirmed). Staff spend time answering routine questions from clients/other departments such as where a container is, when it can be collected, or what its status is.

What we propose.

- A form/portal for clients and internal departments, answering routine questions from existing data.
- Questions it cannot answer, or that need a decision, go to a named person.

How we would measure success (baseline measured first, targets agreed with you):

- Average time from question to answer.
- Staff hours spent on queries per week.

Limitations.

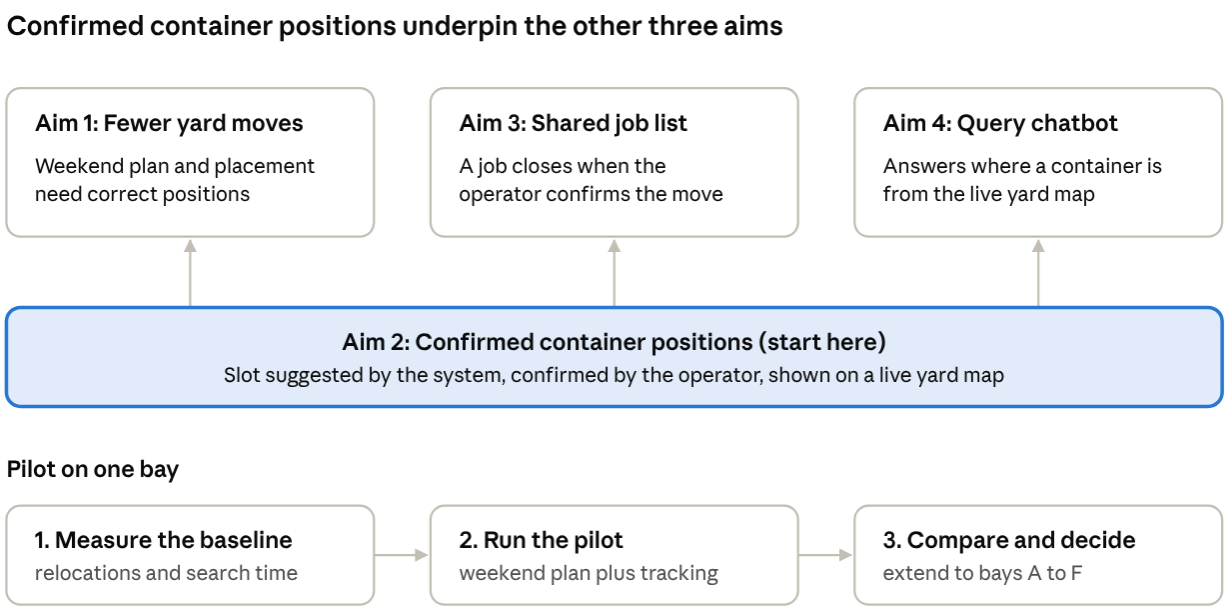
- Its answers are only as accurate as the tracking data (Aim 2).
- It should not make commitments, such as prices or guaranteed times, without a person.
- Each client must see only their own containers.

Key enablers.

- A list of the most common questions and roughly how many arrive per day.
- The channel clients use today (phone, email, WhatsApp, web).
- Who takes over when the form cannot answer the question.
- Your rules on what information each client may see.

Suggested order and pilot

We suggest starting with Aim 2 together with the weekend plan from Aim 1, on one bay, because every other aim relies on correct container positions.



how the four aims depend on each other, and the pilot order

Aim 1 and 2 go first because they are important part of the operation. Aim 3 and 4 follow once container tracking and data collection are in place.

The pilot measures the baseline first, so the before-and-after comparison uses the measures listed under each aim.

Next steps

- [ ] Confirmation and/or correction to this document.
- [ ] NDA so yard and historical data can be shared.
- [ ] Agreement on collaboration structure moving forward.
- [ ] Agreement on the pilot bay and how the baseline is measured.
- [ ] Workflow and coordination method with relevant Cogent staffs.